

SportifyAI – AI-Based Cross-Sport Scout System

1. Problem Statement

Modern sports scouting and athlete performance evaluation require significant manual analysis by coaches and experts. Traditional talent identification methods are time-consuming, subjective, and often fail to detect hidden athletic potential across multiple sports domains. There is a growing need for an intelligent AI-powered system capable of analyzing athlete performance data and sports videos to predict suitable sports, calculate performance scores, and recommend alternative sports based on athletic capability.

2. Existing Solution

Existing sports analysis systems mainly rely on manual scouting, isolated fitness metrics, or basic video analytics. Most existing platforms lack: Cross-sport prediction capability AI-based athlete detection Real-time sports video analysis Automated performance scoring Integrated machine learning recommendations User-friendly intelligent dashboards

3. Proposed Solution

The proposed SportifyAI system combines Machine Learning and YOLOv8 computer vision technology to build an intelligent cross-sport athlete analysis platform. The system can: Predict athlete scout scores Analyze uploaded sports videos Detect sports players and objects Identify sports categories such as Football, Basketball, Volleyball, Swimming, Cycling, and Badminton Suggest alternative sports based on athlete performance metrics Provide an interactive AI-powered dashboard using Streamlit and Flask.

4. Solution Workflow

1. Collect sports videos and athlete datasets
2. Convert videos into image frames
3. Annotate sports objects using LabelMe
4. Convert annotations into YOLO format
5. Split dataset into train and validation folders
6. Train YOLOv8 sports detection model
7. Save trained model as best.pt
8. Train Machine Learning scout score model
9. Integrate AI models into Streamlit/Flask application
10. Display predicted sport, performance score, and recommendations

5. Technologies Used

Python
Streamlit
Flask
YOLOv8
OpenCV
Machine Learning
Pandas
Scikit-learn

6. Advantages

Fast athlete performance prediction
AI-based sports video analysis
Cross-sport recommendation system
Modern interactive dashboard
Automated sports object detection
Real-time intelligent sports analytics

7. Future Scope

Future improvements can include:
Real-time webcam sports detection
Cloud deployment
Mobile application integration
Advanced athlete biomechanics analysis
Deep learning performance optimization
Live sports tracking and analytics

Dataset & Annotation Link

<https://drive.google.com/drive/folders/1H8SFheQFltJFTq9p5lx7FclCOMAZNPZ?usp=sharing>

8. Conclusion

SportifyAI provides an intelligent and automated approach for athlete performance analysis using Artificial Intelligence and Deep Learning. The platform combines scout score prediction, sports video detection, and cross-sport recommendation techniques to assist coaches, athletes, and sports analysts in identifying athletic potential efficiently. This project demonstrates how AI can modernize sports analytics and talent identification systems.